

Paper 1 - Set 13 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Quadratic functions
2	B	Sequences and series
3	C	Statistics
4	B	Exponentials and logarithms
5	F	Differentiation
6	C	Ratio, proportion and percentages
7	C	Differentiation
8	F	Statistics
9	B	Ratio, proportion and percentages
10	C	Functions and transformations
11	D	Exponentials and logarithms
12	D	Integration
13	E	Circles and circle theorems
14	C	Simultaneous equations and inequalities
15	A	Straight lines
16	A	Triangles, radians and sectors
17	D	Circles and circle theorems
18	C	Sequences and series
19	D	Triangles, radians and sectors
20	E	Integration

Worked solutions

Question 1 QUA-12 · Quadratic functions · spec MM1.3

The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .

Answer: C

$$b^2 - 144 = 0, \text{ so } b = \pm 12; \text{ their sum is } 0.$$

Question 2 SEQ-01 · Sequences and series · spec MM2.2

An arithmetic series has first term 5 and common difference 3. Find the sum of the first 40 terms.

Answer: B

$$S_{40} = \frac{40}{2}(2 \cdot 5 + 39 \cdot 3) = 20(10 + 117) = 2540.$$

Question 3 STA-02 · Statistics · spec M6.3

A histogram has a bar for the class $10 \leq x < 15$ of height (frequency density) 3.2 and a bar for $15 \leq x < 25$ of height 1.5. What is the total frequency of these two classes?

Answer: C

$$\text{Frequency} = \text{density} \times \text{width}: 3.2 \times 5 + 1.5 \times 10 = 16 + 15 = 31.$$

Question 4 LOG-15 · Exponentials and logarithms · spec MM5.2

Solve $\log_2(x^2 - 4) - \log_2(x - 2) = 3$.

Answer: B

$$\text{For } x > 2: \log_2 \frac{(x-2)(x+2)}{x-2} = \log_2(x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6.$$

Question 5 DIF-05 · Differentiation · spec MM6.3

For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?

Answer: F

$$f'(x) = 6x^2 - 30x + 24 = 6(x-1)(x-4) < 0 \text{ for } 1 < x < 4.$$

Question 6 RAT-10 · Ratio, proportion and percentages · spec M3.11

£2000 is invested at 5% compound interest per year. Which expression gives the amount after n years?

Answer: C

Compound growth multiplies by 1.05 each year.

Question 7 DIF-18 · Differentiation · spec MM6.3

Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.

Answer: C

$$\text{Derivative } 1 - \frac{4}{x^2} = 0 \text{ at } x = 2; \text{ value } 2 + 2 = 4 \text{ (second derivative } \frac{8}{x^3} > 0, \text{ so a minimum).}$$

Question 8 STA-09 · Statistics · spec M6.3

Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?

Answer: F

$$\frac{1200 + 2100}{50} = \frac{3300}{50} = 66.$$

Question 9 RAT-14 · Ratio, proportion and percentages · spec M3.5

A recipe for 12 biscuits uses 150 g of flour and 100 g of butter. Ann has 1 kg of flour and 500 g of butter. What is the greatest number of biscuits she can make?

Answer: B

Flour allows $\frac{1000}{150} \times 12 = 80$; butter allows $\frac{500}{100} \times 12 = 60$. Butter is the limit: 60.

Question 10 FUN-08 · Functions and transformations · spec MM1.7

For which values of x is $\sqrt{x^2} = -x$?

Answer: C

$$\sqrt{x^2} = |x|, \text{ and } |x| = -x \text{ exactly when } x \leq 0.$$

Question 11 LOG-09 · Exponentials and logarithms · spec MM5.2

Find the value of $\frac{\log_5 27}{\log_5 3}$.

Answer: D

$$\log_5 27 = \log_5 3^3 = 3\log_5 3, \text{ so the quotient is } 3.$$

Question 12 INT-07 · Integration · spec MM7.3

Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

Answer: D

$$a^2 + a = 12 \Rightarrow (a + 4)(a - 3) = 0 \Rightarrow a = 3.$$

Question 13 CIR-15 · Circles and circle theorems · spec MM3.2

A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.

Answer: E

The triangle has a right angle at the origin, so $(6, 0)$ - $(0, 8)$ is a diameter; the centre is its midpoint $(3, 4)$.

Question 14 SIM-04 · Simultaneous equations and inequalities · spec MM1.4

The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

Answer: C

$$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0. \text{ Tangent when } 4k^2 - 8(k^2 - 8) = 0 \Rightarrow -4k^2 + 64 = 0 \Rightarrow k = \pm 4.$$

Question 15 LIN-14 · Straight lines · spec MM3.1

A line passes through $(1, 2)$ with gradient m . It crosses the x -axis at A and the y -axis at B , and the midpoint of AB is $(1, 2)$. Find m .

Answer: A

Midpoint $(1, 2)$ means $A = (2, 0)$ and $B = (0, 4)$. Gradient $= \frac{4-0}{0-2} = -2$.

Question 16 TRI-11 · Triangles, radians and sectors · spec MM4.2

A circle has radius r . A sector of angle θ radians has the same area as the segment cut off by a chord that subtends an angle of $\frac{\pi}{2}$ at the centre. Find θ .

Answer: A

Segment area $= \frac{1}{2}r^2(\frac{\pi}{2} - \sin \frac{\pi}{2}) = \frac{1}{2}r^2(\frac{\pi}{2} - 1)$. Sector area $\frac{1}{2}r^2\theta$. So $\theta = \frac{\pi}{2} - 1$.

Question 17 CIR-02 · Circles and circle theorems · spec MM3.2

Find the complete set of values of k for which $x^2 + y^2 + 2kx - 4y + 3k + 4 = 0$ represents a circle.

Answer: D

$(x+k)^2 + (y-2)^2 = k^2 + 4 - 3k - 4 = k^2 - 3k = k(k-3)$. A circle needs a positive right-hand side: $k < 0$ or $k > 3$.

Question 18 SEQ-11 · Sequences and series · spec MM2.2

The sum of the first n natural numbers is S . The sum of the first $2n$ natural numbers is kS . Which of the following is true?

Answer: C

$S = \frac{n(n+1)}{2}$ and the sum to $2n$ is $\frac{2n(2n+1)}{2} = n(2n+1)$. So $k = \frac{n(2n+1)}{n(n+1)/2} = \frac{2(2n+1)}{n+1}$.

Question 19 TRI-04 · Triangles, radians and sectors · spec MM4.1

Triangle PQR has area $10\sqrt{3}$, $PQ = 8$ and $PR = 5$. Given that angle QPR is obtuse, find QR .

Answer: D

$\frac{1}{2} \cdot 8 \cdot 5 \sin P = 10\sqrt{3} \Rightarrow \sin P = \frac{\sqrt{3}}{2}$, and P obtuse gives $P = 120^\circ$. $QR^2 = 64 + 25 - 80 \cos 120^\circ = 89 + 40 = 129$.

Question 20 INT-11 · Integration · spec MM7.1

The region bounded by $y = 4 - x^2$, the x -axis and the lines $x = 0$ and $x = 3$ is shaded. Find the total shaded area.

Answer: E

From 0 to 2 the curve is above the axis: $\int_0^2 (4 - x^2) dx = 8 - \frac{8}{3} = \frac{16}{3}$. From 2 to 3 it is below: $\left| \int_2^3 (4 - x^2) dx \right| = \left| (12 - 9) - \frac{16}{3} \right| = \frac{7}{3}$. Total $\frac{23}{3}$.